# Migrating to Storage Area Networks: A Report from the Front Line

*An IDC White Paper*

www.idc.com

IDC

*Analyze the Future*

# Migrating to Storage Area Networks

## *A Report from the Front Lines*

*An IDC White Paper*

*Analyst: Eric Sheppard*

### IDC Opinion

*How should an enterprise respond to the increasing demand for high-capacity, flexible, and available storage?*

For many IT organizations, the next area of focus will be the development of consolidated storage area networks (SANs). Central to today's SAN is the switching of Fibre Channel connections between the storage system and application servers. IDC interviewed a sample of current users of McDATA switching products and services. These IT managers reported a variety of benefits enabled by their SANs, including:

- Greater leverage in managing consolidated storage
- Flexible routing of storage services to application servers
- Greater availability of data and the applications that depend on it
- More precise service level agreements within the enterprise

IT managers are encouraged to evaluate SAN technology, which is maturing rapidly. Although SAN is often the architecture of choice for new IT initiatives, IDC also found reports of efficiencies and performance enhancements for thoughtful migrations from existing server-attached storage to SANs.

### Introduction

A storage area network (SAN) is a high-speed network that joins together storage systems and servers. Data is routed between storage systems and servers through a hub or a switch. There are multiple types of hubs and switches, each offering various levels of functionality. Essential to this paper is the fabric switch and the director-class switch (referred to as director). Both switches offer any-to-any (i.e., from storage device to server) connections with dedicated bandwidth. A director is distinguished from a fabric switch by the redundancy within the unit and its design for nondisruptive operation. Once deployed, SANs operate as a utility, making additional storage accessible to application servers as needed, routing data to different servers on demand, and routing data from disk to tape storage for archiving. A SAN is an alternative to traditional server-attached storage.

Today's high-performance SANs utilize a fabric of fiber optic network connections routed by a switch. Bandwidth is often dedicated to guarantee the quality of service. This is done with Fibre Channel fabric switches or directors, which are designed to avoid contention and to minimize latency while providing the SAN with its essential routing ability.

Copyright © 2001, IDC. All rights reserved. No part of this report may be reproduced, stored or distributed via an electronic retrieval system, or transmitted in any form or by any means without express written permission from IDC.

External Publication of IDC Information and Data — Any IDC information that is to be used in advertising, press releases, or promotional materials requires prior written approval from the appropriate IDC Vice President or Country Manager. A draft of the proposed document should accompany any such request. IDC reserves the right to deny approval of external usage for any reason.

SANs address several emerging problems faced by IT managers and provide solutions, such as the need for:

- **Greater overall capacity and flexibility of a unified storage system.** SANs can join together a vast amount of storage resources and make those resources available to an equally vast number of application servers. When storage devices are attached directly to servers, they cannot easily be shared. Adding capacity means taking applications out of service in order to reconfigure the server. For large data centers with hundreds of application servers, the cost of maintaining server-attached storage is high.

- **Better utilization of storage.** When storage is consolidated in a SAN, it can quickly and easily be reallocated. With server-attached storage, designers are forced to provision extra capacity at each server. On a day-to-day basis, this unutilized capacity is stranded. With a SAN, capacity allocation is enterprisewide. As a result, storage managers gain both a higher utilization of available storage and the ability to add storage resources as necessary.

- **Highly available data in support of highly available applications.** Within a SAN, data availability is dependent on the availability of its storage devices as well as its network components (e.g., switches). Today's storage devices can be configured to provide very high levels of availability through redundancy and the continuous operations afforded by hot-swappable disks and power supplies.

- **High-performance (high-speed, low-latency) access to storage.** To be a successful alternative to server-attached storage, a SAN must deliver data to each server without a performance penalty. Today's fabric switches provide up to 1Gbps port speeds with latencies of less than 2 microseconds. Servers are free to apply more processing cycles to the application because storage and network management is provided by the SAN's infrastructure components and their related software.

- **More efficient management of storage.** SANs provide a platform for storage management software that assists technicians in maintaining the system. At the fabric level, operators can study usage patterns and optimize the use of storage resources. For example, storage resources configured for high availability can be routed to mission-critical applications, while less critical applications can be supplied with less costly storage resources.

- **Flexibility to grow and scale storage.** SANs can seamlessly integrate additional storage and provide data services to increasing numbers of servers. Adding a switch can increase the network capacity of the SAN. Zoning techniques allow the SAN to be partitioned for management or security purposes.

Many current adopters of SAN technology are reporting success. In March 2001, IDC spoke with customers of McDATA, a supplier of SAN directors, switches, software, and professional services. Our sample of McDATA customers included enterprises in different industries and with different system requirements. Reports from these companies, detailed below, contain a mixture of common benefits (e.g., increased ease of maintenance) and unique benefits (e.g., the ability to decouple storage requirements from application design).

### *Growth in Enterprise Data Drives SAN Adoption*

The fundamental force accelerating SAN adoption is the rapidly increasing amount of data that enterprises collect and utilize to increase competitive advantage. Not long ago, enterprise data consisted of only numeric or textual data types. Today, new data types, such as image, audio, and video data, are increasingly common, and their storage requirements are immense.

The explosion in critical data affects many industries. In healthcare, for example, medical practitioners now commonly store EKG, X-ray, and other diagnostic data electronically and distribute it to remote diagnostic services. In the pharmaceutical industry, a shift in research methods is underway as the human genome becomes a navigational model for rational drug design. The industry anticipates a tenfold increase in the number of compounds to be tested in the coming years. In the future, medical records will be electronic and will include a genetic map for each individual.

Inspired by integrated enterprise resource planning systems, enterprises are now keenly aware that data only has value when it can be put to work across the enterprise and even outside traditional enterprise boundaries. Customer data, for example, is critical not only to fulfillment, billing, and accounts payable systems, but also to marketing, strategic planning, and inventory management.

Supply chain integration initiatives extend data sharing outside the enterprise as well. Replicates and aggregates of operational data are shipped to trading partners so that suppliers can optimize production schedules and distribution strategies. From a data storage point of view, supply chain integration means that information once stored uniquely is now being cached in other locations.

## Scalability Is a Top SAN Benefit

Many enterprises are discovering that the cost of maintaining server-attached storage is growing to prohibitive levels and this high cost is effectively limiting storage scalability. SANs provide the opportunity to centralize storage and gain economies of scale, particularly with respect to management.

In a study of consolidated storage, IDC found that a data administrator could manage about 7.5 times as much data when it is centralized. After normalizing to adjust for the higher salaries of skilled data administrators, IDC concluded that maintenance costs for consolidated storage are reduced by a factor of 3.75 when compared with the costs of maintaining distributed, server-connected storage (*The Pursuit of Excellence: Harnessing the Power of Change in the Data Center*, IDC #20662, October 1999).

In addition to reducing maintenance, SANs provide an architecture where additional storage can be brought online without disrupting ongoing operations. Storage devices can be installed or upgraded by linking to the SAN fabric and put into immediate use. All of this activity can be isolated entirely from server operations. Furthermore, any-to-any connectivity allows storage managers to allocate storage as needed and to reallocate storage as current projects end and new ones are launched.

## High-Performance SANs Depend on Fast Fabric Switching

The foundation of a SAN is its network-switching technology. Bandwidth, latency, number of ports, and their degree of availability differentiate SAN switches.

Designed specifically for storage, Fibre Channel's 1Gbs bandwidth and low latency offer an ideal transport for supporting data transfer rates that meet even the most demanding server requirements. With 2Gbs switches coming shortly and 10Gbs switches quickly moving off the drawing board, Fibre Channel's bandwidth is

### New Data Requirements for a Major Bank

IDC spoke with the director of storage systems services at a major bank. Enterprise data storage management became a director-level position six weeks ago as the bank continued on its path to consolidated data storage. With 15 mainframes and 6,000 servers, the task of unifying storage has been a substantial undertaking. Tape, optical, and magnetic disk storage systems are back-end elements of the bank's SANs, which are under construction at two sites that are several hundred miles apart.

McDATA directors were central to this company's 500-port fabric, and McDATA switches (ES-1000) provide 4-1 fan-in for the bank's 100 tape drives. There are currently 65TB of storage contained in the SAN. Second-tier data centers are being consolidated at this time as the SAN continues to grow.

"I need the SAN in part because I am encountering large new storage requests," IDC learned. "For example, we are planning to bring on board a check imaging system. We will process 17 million checks per month, and each scan generates about 35KB of data. That means that we'll be storing 600GB per day. We will hold the data on magnetic disks for 7 to 63 days, depending on the check, which we estimate will require 58TB of disk. We will then move images to a tape archive that we estimate to be just over a petabyte in size. And this is just one new application that our SAN must support. I expect more requests like this one, for example, for voice message storage."

### Storage on Demand at a Major Telecommunication Service Company

IDC spoke with the director of storage systems for a major diversified telecommunications services company. The company is in the process of deploying SAN technology, a project started 18 months ago. Successful pilot testing led to a commitment to consolidate storage services for the company. Anticipated increases in demand for storage capacity also drove the decision to build a SAN infrastructure.

McDATA directors provide fabric switching for the SAN, which had 784 ports supporting production systems in March 2001. With this month's addition of 20TB of additional storage, the SAN provides access to over 150TB of disk storage. Archival tape systems are also incorporated into the SAN.

"Technical planning is key to a successful SAN implementation," IDC was told. "Anyone planning a major SAN implementation would be wise to focus attention on designing the correct fiber management infrastructure. This involves the physical layout of the data center, giving thought to the placement of storage resources, the relationship of fiber cabling to storage system zoning, and the placement of switches."

"The efficiencies of SAN are very real. We find that the acquisition cost of provisioning storage is 35-40% lower now that our SAN is in production. In addition, the shared, scalable SAN environment allows the team to efficiently manage 150TB of storage with just 19 people."

**SANs Enable Service Level Agreements at a Growing ASP**

IDC spoke with Bill Augustadt, Chief Architect for eOnline, Inc., a company founded in 1999 that is an application service provider (ASP). ASPs provide storage, servers, and application management for their customers, who choose to contract for IT services rather than host them internally. Six SANs provide the basic infrastructure for the company, which hosts applications such as SAP for hundreds of customers. This company did not migrate to the SAN architecture but rather designed its business with SANs from the onset.

McDATA directors are at the core of the company's SANs, and McDATA switches support user access at the edges. When the company began planning its infrastructure, SCSI cabling was considered, but this plan was quickly abandoned due to the sheer number of cables. Higher capacity fiber fabric was clearly a better approach. The SANs at each of six data centers have 196 ports in production, and the company currently plans to grow each SAN to 256 ports. With the arrival of 64 port directors, the company plans to move on to 512 ports per data center.

"Our business is based on service level agreements (SLAs), and if our systems aren't up and running, then we're not in business." IDC learned. "One of our customers estimates the cost of downtime to be $100,000 per half hour. Our business absolutely depends on our ability to provide customer access to their applications. We must provide multiple, redundant systems that can be maintained and managed easily."

"We believe that we manage our infrastructure with one-third the staff that would otherwise be required and, in conjunction with the need for high availability to underpin our SLAs, we think we have a highly cost-efficient solution."

expected to grow in concert with (if not faster than) the most demanding applications.

Today's enterprise SANs typically have dozens or hundreds of ports, with thousand-port systems clearly visible on the planning horizon. This creates the technical challenge of avoiding contention as hundreds of servers call upon the unified data stored in the SAN.

For large SANs, there is a significant advantage to using larger fabric switches and directors as core building blocks. By using directors and fabric switches with a large number of ports, SANs can grow without losing an exorbitant number of these ports as interswitch links (ISLs). As companies begin to scale a SAN, they will quickly discover that these ISLs are unavoidable and will ultimately consume some of the available ports. For example, two 64-port directors could require 8 ISLs and thus provide only 120 ports available for storage and server devices. Four 32-port directors could require as many as 16 ISLs and provide only 112 ports. Eight 16-port switches could use 24 ISLs leaving only 104 ports. As these examples suggest, switches or directors with 64 or more ports result in a SAN that is simpler in design and easier to manage.

SANs that are intended to ensure maximum application uptime are designed to be highly available. Directors provide high availability with redundant, integrated components resulting in a single, highly-available product. Fabric switches achieve high-availability when redundant units are deployed. Redundant design within directors provides 99.999% availability at the center of the SAN. Of the available switches at the edge of the SAN, 99.99% can be replicated as needed at the device level.

## Interoperability Is Critical to SAN Infrastructure

Multivendor IT environments are the rule in today's enterprise, and SANs are no exception. Interoperability is most critical at the SAN switch, which interoperates with hardware and software provided by other suppliers.

Interoperation between SAN switches and storage devices is well understood. Alliance programs that ensure systematic testing of components, such as the Fibre Channel Industry Association, assure that the "back end" of the SAN works properly.

The server side of the SAN presents greater interoperability challenges. Each server is equipped with a host bus adapter (HBA) that provides connectivity to the switch. Different server operating environments require different drivers to support the HBA. Operating system and application software must be configured correctly in order for the SAN to provide data to the servers.

SANs that include switches from different suppliers present the greatest interoperability challenge. While Fibre Channel standards enable the switches to move data among switches from different vendors, variances in firmware make it difficult to gain a fabric view of the SANs today. That said, the industry continues to move toward interoperability standards.

### Improved Performance for a Data Warehouse Company

IDC spoke with the senior technologist for a company that cultivates a data warehouse of industry information and sells access to that data to members of the industry. The company was one of the first to field a SAN solution driven by capacity constraints encountered with a traditional server-attached architecture.

Six McDATA directors provide fabric switching for the company's SAN, and McDATA engineers participated in the design of the system. Sun servers support Oracle databases and a collection of analytic tools. The system supports a relatively small collection of 20–30 power users who issue complex queries against the databases.

The primary motivation to build a SAN was to improve input/output (I/O) performance for the data-intensive operations of the data warehouse. The company wanted to decouple storage purchases from server purchases in order to optimize each system separately.

"We're not a 7 x 24 company, we're a 5 x 16 company, and a SAN still makes good sense for us," IDC learned. "While our SAN has been quite reliable, we built it primarily to improve the performance of our data warehouse. Our need is for flexible access to large data sets."

"McDATA has been a good citizen in a system that incorporates hardware and software from many different suppliers. Interoperability makes it possible for us to reduce cost by shopping for exactly the components we need."

## Migration to a SAN Takes Time and Requires Planning

SANs are not always designed and implemented with entirely new servers and storage systems. Quite often, IT managers realize that server-attached storage is inefficient and begin the planning process for migrating to a SAN. A pilot SAN is often constructed, perhaps in support of a new application, and tested prior to the migration of existing production systems. For large IT installations, the migration may take months.

An inventory of current application, server, and storage resources is the first step. Not all applications are equally important to the enterprise, and consolidated storage has a varying effect on different applications. A report writer may need access to wide varieties of data, for example, that need not be highly available. A transaction system, on the other hand, may demand 24 x 7 service only for a well-bounded set of data.

Migrating to a SAN may also enable IT staff to address other issues. Data centers may be consolidated, for example, in order to enhance the cost savings. High-speed linkages between sites may be included in the SAN migration plan in order to address disaster recovery objectives. With enterprise-wide data consolidation, new applications may be considered that bring together information once held in separate, distributed systems.

System design, interoperation, and migration expertise are increasingly available to assist in SAN migration. Many vendors maintain test laboratories to make certain that new models of product continue to interoperate as

### Migration Planning at a Fortune-10 Consumer Products Company

IDC spoke with a technology consultant specializing in the SAP and storage from a diversified consumer products corporation. The company has acquired several brand names over the past years and its information-processing needs must be integrated into an enterprise strategy. A SAN has been in the planning and design process over the past year, and the first switching devices for a pilot system have just arrived to begin implementation.

Two McDATA 32 port directors will be used for the initial implementation of the SAN. In addition, McDATA engineers have participated in the design process. The SAN will eventually aggregate existing server-attached storage as well as become the platform upon which future storage capacity will be added. The company is also consolidating data centers to two primary centers separated by several hundred miles.

"SAN technology is complex, and McDATA's assistance in the design process was crucial," IDC learned. "I can't be exact, but I suspect that, left on our own, planning would have taken twice as long and may well have included errors due to our inexperience. McDATA's experience and its System Integration Laboratory has been helpful to us."

"We have a wide array of legacy systems that primarily run SAP applications. We have inherited IT systems and storage facilities as a part of the corporation's acquisition of new brands. SAN is a way for us to bring disparate systems into a common storage architecture and manage storage more effectively.

"An important part of the planning process was preparing an inventory of existing applications along with their processing, storage, and availability requirements. That inventory makes it possible for us to prioritize and plan the order in which applications are moved to SAN-attached storage."

**Greater System Resilience at a Fortune 100 Financial Services Company**

IDC spoke with the manager of technology planning and infrastructure support for a division of a major financial services company. The division has three service centers in production and a fourth under design. Over 350 servers manage $6 trillion in funds. Planning for a SAN began 18 months ago as concerns over server-attached storage emerged.

The primary motivation for shifting to a SAN architecture was to provide remote data-copy functionality between different cities in order to improve disaster recovery capabilities. Other motivations included a concern about a growing amount of space in data centers devoted to disks and the increasing cost of managing distributed disk storage.

McDATA current directors and switches provided the high-port density that the company's data center initially required. Key to choosing McDATA technology was its technology road map pointing to today's 64-bit directors. To manage increasing storage capacity, higher fanout is required. Chaining switches with smaller port counts are more complex, and the resulting system is less reliable.

"When someone bought too much disk with a server, it just sat there unused, and when someone else bought too little disk, we needed to rush to add capacity," IDC learned. "We looked to SAN technologies as a better way to manage storage resources. We can now reallocate storage to where it is needed. This is one of the basic benefits of SAN."

"We are piloting remote data copy right now, and the numbers are encouraging. We expect to improve our system resiliency as we are systematically able to back up critical systems using the SAN as a foundation."

they should. Component suppliers, system suppliers, and system integrators provide consulting services aimed at streamlining the planning process and identifying potential problems based on the experience of early adopters.

## SANs Enable Strategic Planning for Storage

For large data centers, managing server-attached storage has presented increasing tactical challenges. Upgrading hundreds of servers to support shifting enterprise application needs has kept much of the IT staff's attention on immediate problems.

The shift to a consolidated SAN architecture enables IT managers to take a longer, more strategic view. Through the use of service level agreements (SLAs) between the IT staff and enterprise information users, storage planners can align their SAN architectures to match corporate objectives and anticipate growth in storage needs. Moreover, the SAN promotes the ability to provide both high-speed, highly available storage services as well as lower-performance, lower-cost service when appropriate.

With SANs, a new utility storage model is emerging allowing application developers to focus their attention on application functionality without the need to reengineer storage infrastructure with each new project. Just as electrical power and network services are delivered to applications on demand, so too are storage services.

## Challenges to Delivering Value with SANs

SAN technology is not static, and there remain challenges to users and suppliers alike. IDC's research uncovered three key concerns voiced by SAN users.

- Multiple standards exist. For some interfaces a single standard dominates (e.g., Fibre Channel). For others, there remain choices and attendant tradeoffs that are not fully understood (e.g., block level data over IP, which are expected to raise performance issues).

- For enterprises contemplating a shift to SANs, the justification may be difficult to articulate, particularly if systems are operating reliably today. The IT managers that IDC spoke with remarked that in some cases, the shift to SAN was in anticipation of problems only just beginning to emerge.

- The importance of linking application functionality to enterprise objectives is increasingly clear when storage systems become a utility infrastructure. The SLA is more than a contract between divisions of the IT department. High-quality SLAs reflect the priorities that enterprises set forth for different categories of information.

SANs can help improve the strategic alignment of IT by streamlining application development, reducing the cost of storage, increasing the flexible provisioning of storage, and thus freeing developers to focus more exclusively on an application's value to the enterprise.

## Summary and Conclusion

IDC's research confirms that the move to consolidated storage is underway. Further, though still a relatively young architecture, current SAN deployments are providing real benefits to real companies. Three major benefits include:

- **Manageability at the fabric level.** A SAN central switch or director provides the platform for monitoring and provisioning tools that enable technicians to manage a vast amount of storage resources.

- **Availability up to 99.999% level.** SANs deliver highly available storage services to applications through multiple routes, thus eliminating single points of failure that have plagued earlier architectures. Those surveyed used directors to reach 99.999% availability.

- **Scalability of storage services.** SANs provide the basis for a cost-effective, shared storage model supported by service level agreements among the key stakeholders within the enterprise.

As the examples above suggest, reports from our sample of IT managers directly responsible for SAN deployment indicate that the benefits associated with SANs are often demonstrable, and rarely insignificant.

2001

*IDC delivers accurate, relevant, and high-impact data and insight on information technology to help organizations make sound business and technology decisions. IDC forecasts worldwide IT markets and adoption and technology trends, and analyzes IT products and vendors, using a combination of rigorous primary research and in-depth competitive analysis. IDC is committed to providing global research with local content through more than 500 analysts in more than 40 countries worldwide. IDC's customers comprise the world's leading IT suppliers, IT organizations, and the financial community. Additional information on IDC can be found on its Web site at http://www.idc.com.*

*IDC is a division of International Data Group, the world's leading IT media, research, and exposition company.*

www.idc.com

IDC

*Analyze the Future*

5 Speen Street • Framingham, MA 01701
(508) 872-8200 • Fax (508) 935-4015
www.idc.com

*01-137STORAG*
*Sponsored by McDATA*
*April 2001*